3. The Real Estate Track - Keppel

Description:

The **Real Estate Track** features real-world business challenges contributed by **Keppel Group**. *Keppel's real estate division is an innovative urban space solutions provider that leverages technology to deliver sustainable, customer-centric solutions.*

Participants will tackle industry-relevant problem statements and leverage AI technologies to develop practical, impactful solutions for the banking and financial services sector.

Requirement:

This challenge presents **ONE** real-world enterprise challenge statements.

Each team is required to select only **ONE** case study to solve.

Please clearly indicate your chosen case study at the beginning of your presentation.

Challenge Statement:

AI HARVEST – Turning Expert Know-How into Reusable Intelligence

Introduction

Modern commercial real estate assets depend on knowledge across many functions. Experienced professionals use judgement, intuition and accumulated experience to help assets operate efficiently, serve customers and create long-term value. Commercial real estate assets may include office towers, logistics facilities and data centres, involving engineering, maintenance, energy, leasing, tenant services, sustainability, safety and operations.

The business issue is that this know-how is not always easy to transfer or use consistently across different people and assets:

- Critical knowledge may be concentrated among a limited number of experts.
- New employees may require significant time to build the required expertise.
- Best practices may not be applied consistently.
- Institutional knowledge may be lost when experienced employees retire or move on.

Problem Statement

How might an organisation systematically turn human expertise into reusable intelligence that supports decisions, operations and continuous improvement across a portfolio of assets?

Challenge

Develop a platform and operating model for turning expert know-how into reusable, governed domain capabilities called “Intelligence Pills”. Each pill represents a defined area of expertise and may incorporate relevant knowledge, heuristics, decision logic, guardrails and escalation requirements. Teams are encouraged to focus on a specific scenario from below intelligence pills:

- Asset Operations
- Energy Optimisation
- Leasing
- Technical Services
- Sustainability
- Tenant Experience

What the Solution Should Solve

- **Capture expert know-how -** Provide a way to collect knowledge from experienced professionals, including knowledge that may not already be documented.
- **Turn know-how into reusable, governed intelligence -** Structure the captured expertise as a reusable domain capability with defined context, decision logic, guardrails, ownership and approval requirements.
- **Apply specialist AI support -** Apply the relevant domain capability through one or more specialised AI agents or embedded workflows; a pill need not correspond one-to-one with an agent.
- **Coordinate across domains -** Allow different specialist agents to work together when an issue spans more than one area.
- **Keep people in control -** Enable people to guide, supervise and review the AI-supported process and recommendations.
- **Learn through governed feedback -** Capture operational outcomes, expert corrections and new experience for review, validation, versioning and approval before release into operational use.

The Solution Should Be

- **Human-supervised** – Show where a person guides, reviews or decides, rather than allowing the AI to act without clear supervision.
- **Transparent** – Make it understandable how the solution reached a recommendation or what knowledge it relied on.
- **Accountable** – Clarify who is responsible for using, reviewing and improving the capability.
- **Reliable** – Explain how recommendations would be checked and kept dependable.
- **Reusable** – Show how the knowledge or capability can be used again for similar needs.
- **Scalable** – Explain how the approach could support multiple asset types and areas of expertise.
- **Governed** – Describe how feedback and new expertise would be captured, reviewed, validated, versioned and approved before release.
- **Portable** – Explain how the organisation retains ownership of captured knowledge, decision logic, derived skills and feedback history, including transfer independent of a particular vendor or model.
- **Version-controlled** – Show how approved knowledge and domain capabilities would be versioned, updated and, where necessary, rolled back.
- **Secure** – Identify how access to operational data and expert knowledge would be controlled according to role, sensitivity and intended use.
- **Clearly bounded** – Define what the solution may recommend or execute, what requires human approval, and what must be escalated because it falls outside the approved scope.

The Solution Should Include

Demo Walkthrough – A live demonstration.

Solution Diagram – Solution should clearly illustrate:

1. The specific business problem being addressed.
2. Whose expertise is being captured and how knowledge that is not already documented would be obtained.
3. How the captured expertise becomes reusable rather than remaining a one-off answer.
4. How an Asset Operations Manager would use and supervise the solution, verify its recommendations and understand when human escalation is required.

5. Why the proposal is more than a chatbot or information-retrieval tool.

6. How feedback, outcome data or new expertise would be reviewed, validated, versioned and approved before incorporation after deployment.

7. How the concept could extend from one asset or domain to multiple assets, business units or enterprise environments.

8. A credible path to implementation using current or near-term technology.

Source Code – Complete source code submitted through a GitHub repository.

Note: *The features listed above are provided as guidance only. Participants are strongly encouraged to explore alternative approaches that meaningfully address the problem statement.*

7. Suggested Tools

Tencent Cloud WorkBuddy Ecosystem

The following tools are all accessible through WorkBuddy's built-in ecosystem — no complex development environment setup required. Operate everything via natural language.

Area	Recommended Tools	Description
Agent Development	WorkBuddy	• An AI-native workspace that enables users to build, deploy, and manage intelligent agents and automations through a natural-language interface, integrating large language models, MCP connectors, skills, and scheduling to automate real-world tasks across business workflows.
Agent Development	CodeBuddy	• Tencent Cloud's AI coding assistant that provides intelligent code completion, code review, debugging, and multi-file editing capabilities within the developer's IDE, accelerating software development with context-aware suggestions.

Tencent Cloud Services Recommended for Hackathon Challenges

Area	Recommended Tools	Description
Agent Development	Tencent Cloud Agent Development platform(ADP)	• Tencent Cloud's foundational infrastructure for building AI agents at scale, offering model orchestration, sandboxed runtime environments, tool-calling frameworks, and extensible plugin ecosystems that allow developers to compose multi-agent systems with guardrails, RAG, and human-in-the-loop controls.
Visual Asset	Miora	• An AI-native creative studio that generates production-grade visual assets – including images, videos, 3D elements, and user interfaces – directly from natural language descriptions.
Agent runtime sandbox	Tencent Cloud Agent Runtime	• Agent Runtime uses a secure sandbox as its core execution environment, supporting millisecond-level startup and concurrency of tens of thousands of instances. It provides a secure, isolated, and high-performance execution foundation for AI Agents.
Speech & Voice	ASR from Tencent Cloud TRTC	• Real-time and batch speech-to-text recognition with multi-accent support, used to transcribe spoken transaction commands into text for intent parsing.
Speech & Voice	TTS from Tencent Cloud TRTC	• Natural text-to-speech synthesis for voice-based clarification prompts and confirmation overlays in the conversational banking experience.
Big Data / Database / Storage / Compute / Container / Network	Tencent Cloud	• Tencent Cloud provides the full underlay infrastructure.

8. Project Submission Requirements

Project Basic Requirements

The project must be original and built on at least one of the products CodeBuddy or WorkBuddy.

Proof of product usage is mandatory: chat screenshots, API call logs, or a written development-process description. Without proof, the project will not proceed to scoring.

Submission Requirements

Submission Items	Req.	Description
Project title	Required	• The name of your AI Agent project
Short blurb	Required	• A summary of what your project does or the value it delivers. Hard limit: under 10 words
Project Description	Required	• Project Overview: Target Scenarios, Users, and Value Proposition • Real-World Scenario Insights: Source of Pain Points, Target Audience, and Core Problems Solved • Comprehensive Solution Design: business and technical architecture, and how prompts drive the AI generation. • Business Value: quantifiable metrics or clearly defined impact.
CodeBuddy / WorkBuddy Conversation History	Required	• The CodeBuddy / Workbuddy chat history used during the project development process
Cover Image	Required	• A 16:9 cover image for your project, used for the online showcase. Recommended size: 380×216px.
Demo video	Optional	• A 5–8 minute video covering: • Project overview • Core Agent features and how it's used • A short reflection on your build approach and any development-tool tips
Chat history	Required	• Minimum of 3 screenshots of your chat logs from CodeBuddy or WorkBuddy during the development process
Project link	Optional	• A live URL or demo link for your project. Optional, but earns bonus points.
Other Requirements as per the Challenge Statement	Optional	• Please check the respective challenge statement for specific requirements.

9. Competition Timeline

16 Sept 2026: Challenge Kick-off

Challenge launch & submissions open

16 Sept - 16 Oct 2026: Online & Offline Training

Hands-on workshops

16 Oct 2026: Project Submission

23 Oct 2026 : Finalist Announcement

Shortlist Top 2 teams per track for Demo Day

3 Nov 2026 (TBC): Singapore Hackathon Demo Day

Crown the Winners

10. How to Participate

Step 1: Register

Sign up via the registration link below. Only **one registration** per team is required. Each team may consist of **1–3 members**, and all participants must be based in Singapore.

Registration Link: https://qdrl.qq.com/65xN1yft

Step 2: Join the WhatsApp Group and Receive the Hackathon Handbook

You will be invited to join the respective track's **WhatsApp group**, and the Hackathon Handbook will be sent to you via email.

Step 3: Create Your Accounts and Claim Your Credits

Each team member may create CodeBuddy, WorkBuddy, and Miora accounts to receive complimentary credits and access Tencent Cloud's AI-powered development tools.

Step 4: Build and Submit Your Project

Develop your project using CodeBuddy, WorkBuddy, or Miora, and submit it before the deadline.

Project Submission Link:
https://tinyurl.com/TCHackathonSGProjectSubmission

11. Judging Process

- **Preliminary Technical Judging:** Each project will be evaluated by **the respective industry organization that contributed the challenge statement** and **Tencent Cloud experts**. As each track has different objectives and requirements, the detailed judging criteria will be shared with participants after registration.
- **Demo Day Grand Final:** Representative teams from participating schools present their projects live at the offline roadshow. Judges score each project based on theme alignment, use of AI tools, and game quality, and select regional award winners and teams advancing to the grand final, please refer to the following:

Evaluation Dimension	Score	Key Review Focus
Impact & Relevance	10 Points	Evaluate whether the project addresses a real-world problem and creates meaningful value.
Human-Centered Design	10 Points	Assess how well the solution is designed for real users.
AI Interaction	10 Points	Evaluate the quality and depth of AI usage in the project.
Technical Execution	10 Points	Assess the technical quality and completeness of the project.
Feasibility	10 Points	Evaluate whether the project is realistic and scalable beyond the hackathon.
Demo & Storytelling	10 Points	Assess how effectively the project is presented and communicated.
Innovation & Creativity	10 Points	Evaluate the originality and uniqueness of the project.
User Experience (UX) & Accessibility	10 Points	Assess the overall usability and accessibility of the solution.
Responsible AI & Ethics	10 Points	Evaluate whether the project demonstrates responsible and trustworthy AI practices.
Overall Quality & Judge's Impression	10 Points	Provide an overall assessment of the project.

11. Participant Benefits and Awards

Registration Benefits - For All Participants

Track tool	Credit Allocation
CodeBuddy/WorkBuddy	1,000 credits / person

Grand Final Awards

The top 2 teams from preliminary judging will present at the Grand Final Demo Day on 3rd November 2026 , competing for the grand prizes below.

Award	Team Quota	Prize
First Prize	1	SGD $10,000
Second Prize	1	SGD $5,000
Third Prize	1	SGD $2,000

Additional Benefits

Benefit	Description
Internship Fast Track	Outstanding participants may enter the fast-track final interview process for Tencent CSIG internships
POC, Investment & Incubation Opportunities	Outstanding projects may have the opportunity to pursue proof-of-concept (POC) collaborations and receive investment and incubation support from our VC partner.
Official Certification	An official Tencent Cloud certification certificate with a unique online verification code and lookup entry

13. Terms and Conditions

Intellectual Property and Licensing

The intellectual property rights of the entries belong to the individual entrant or team. Entrants must guarantee that their entries are original and do not infringe upon any third-party rights. Entrants grant the organizer (Tencent Cloud) a non-exclusive, royalty-free license to use the project name, description, demo screenshots / screencasts, and team information for non-commercial purposes, such as challenge promotion, case studies, and media coverage. The organizer will credit the project and team when using such materials.

Privacy and Data Statement

The Organiser will collect and process the personal information submitted by participants (such as name, contact details and team information) . Such information will be used solely for the purpose of event registration, judging, notifications, and related publicity or follow-up, and will not be used for unrelated purposes or disclosed to any unauthorised third party. Participants may request access to, correction of, or deletion of their personal information via the official contact channels.

All the best for the Challenge — AI CAN DO IT !

Tencent Cloud | AI SINGAPORE | CODEBUDDY | WORKBUDDY | Miora Tencent Design

We look forward to receive your projects!

Register Now!

